

Crime Analysis Review (2023 – 2026)

UK Regional Dynamics, Socio-Economic Drivers & Strategic Recommendations

Scope & Data Framework

- ✓ **Temporal Scope:** 3-year continuous timeframe spanning from June 2023 to May 2026.
- ✓ **Population Standardization:** Crime metrics calculated per 1,000 residents using ONS population data to compare urban and rural authorities fairly.
- ✓ **Geographic Resolution:** Evaluated across Lower Layer Super Output Areas (LSOAs) and Local Authority Districts (LADs).
- ✓ **IMD Decile Mapping:** Index of Multiple Deprivation deciles (1 = Most Deprived 10%, 10 = Wealthiest 10%) incorporated across 7 key social domains.

Data Notes & Context

- **Deduplication:** Data pre-processed via `dataset.drop_duplicates()` to ensure single incident counting.
- **Log Transformation:** Applied $\log_{10}(\text{rate per 1000})$ to handle right-skewed crime rates when conducting linear regression against IMD deciles.
- **IMD Domain Weighting:** IMD captures Income, Employment, Health, Education, Housing, Crime, and Living Environment.
- **Per Capita Benchmark:** raw count toggles are available alongside per-capita metrics for high-density areas like Westminster.

Presentation Agenda



1. TRENDS & SEASONALITY

Macro analysis of monthly rates & seasonal peaks



2. VIOLENCE & SA

Deep dive into violent crime category trajectories



3. DISTRICT ANALYSIS

Comparative frameworks for individual local authorities



4. IMD DECILES

Regression modeling on deprivation impact

Interactive Dashboard Demo

Switching to Power BI for dashboard demo.

Overview of Findings

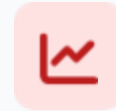
Analysis Pillar	Observed Outcome	Strategic Implication
Trend & Seasonality	Slight downward trend with predictable winter/summer peaks.	Enables proactive seasonal officer shift planning.
Violence & Sexual Assault	Specific upward trends identified in Surrey and Greater London LSOAs.	Requires targeted intervention.
IMD Deprivation Impact	Strong linear regression correlation explaining ~45% of crime rate variance ($R^2 = 0.453$).	Suggests that socio-economic status is the primary underlying crime driver.
Hotspot Districts	Westminster exhibits severe non-violent outliers (Theft)	Driven by high nighttime economy density rather than resident population.

Strategic Next Steps & Recommendations



1. Targeted Relief for Lower-Income Districts

Focus social intervention programs and community support funding on districts within **IMD Deciles 1–3** to mitigate deprivation-driven crime drivers.



2. Advanced ARMA Predictive Modeling

Expand analysis beyond descriptive regression by deploying an **ARMA time-series model** to forecast short-term crime rate shifts based on historical lag.



3. Westminster Nighttime Patrol Surge

Increase police presence and body count in **Westminster during nighttime hours** to counter elevated theft and drug offenses linked to the night economy.



4. Targeted Violence & SA analysis

Establish a dedicated investigation task force to determine the specific root causes driving rising **Violence & Sexual Assault** rates in **Surrey and London**.

Questions

Thank you for your time!